
MODIFY
DIVE!
LOW RISK
Novice divers proceed
HIGH RISK
expert divers only
ASCVDID or higher
MALLOW
DIPPIN'S
long lasting flavors!
Calci-Yum
Collar Yum
MAC
BASIC
SWITCH
Angel of New Orleans
80
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60
61
62
63
64
65
66
67
68
69
70
71
72
73

5.2 - HTN Management

1) 1° Submarine leaking high pressure bubbles:
Primary HTN accounts for 95% of all cases of hypertension

2) Depth marker with 20mm increments:
Every 20 mmHg increase in SBP increases risk of cardiovascular death by 2x

3) 2 risky angler fish:
Every 20 mmHg increase in SBP **increases risk of cardiovascular death by 2x**

4) Anxious elderly lady:
Consider **less strict BP control for elderly patients** >75 with many comorbidities

5(P) MODIFY tank:
Address **lifestyle modification** with exercise, weight loss, the DASH diet, potassium supplementation, limiting salt intake, & limiting alcohol intake **(2)**

6(P) Chubber diver swimming:
Address lifestyle modification with **exercise**, weight loss, the DASH diet, potassium supplementation, limiting salt intake, & limiting alcohol intake **(2)**

7(P) Loose belt around chubby diver:
Address lifestyle modification with exercise, **weight loss**, the DASH diet, potassium supplementation, limiting salt intake, & limiting alcohol intake **(2)**

8(P) DASH Diet cookbook:
Address lifestyle modification with exercise, weight loss, **the DASH diet**, potassium supplementation, limiting salt intake, & limiting alcohol intake **(2)**

9(P) Bananas on shelf:
Address lifestyle modification with exercise, weight loss, the DASH diet, **potassium supplementation**, limiting salt intake, & limiting alcohol intake **(2)**

10(P) Broken salt shakers:
Address lifestyle modification with exercise, weight loss, the DASH diet, potassium supplementation, **limiting salt intake**, & limiting alcohol intake **(2)**

11(P) Broken booze bottles:
Address lifestyle modification with exercise, weight loss, the DASH diet, potassium supplementation, limiting salt intake, & **limiting alcohol intake (2)**

12) Gauge on diver:
Stage 1 hypertension = SBP 130-139 mmHg or DBP 80-89 mmHg

13(P) “HIGH RISK” sign on gunky, red atherosclerotic pipe:
Patients with **Stage 1 HTN** **& HIGH (≥10%) risk for atherosclerotic cardiovascular disease (ASCVD10)** should receive **antihypertensive medications** in addition to lifestyle modifications **(3)**

14(P) “LOW RISK” sign pointing to lifestyle modifications:
Patients with **Stage 1 HTN** **& low ASCVD10 risk (<10%)** should receive **lifestyle modifications (4)**

15(P) Diver with white CHLORO thighs:
For stage 1 HTN, options for initial pharmacotherapy include long-acting calcium channel blockers, **thiazide diuretics**, ACE inhibitors, or ARBs **(5)**

16(P) Broken Calci-Yum soft serve machine:
For stage 1 HTN, options for initial pharmacotherapy include **long-acting calcium channel blockers**, thiazide diuretics, ACE inhibitors, or ARBs **(5)**

17(P) Cracked ace of spades:
For stage 1 HTN, options for initial pharmacotherapy include long-acting calcium channel blockers, thiazide diuretics, **ACE inhibitors, or ARBs (5)**

18) Diver with white thighs & CHLORO underwear:
Thiazide-type diuretics (e.g., chlorthalidone, HCTZ) are first-line treatment options for primary HTN

19) Treasure chest of complications:
Thiazide diuretics can cause a variety of adverse effects (e.g., hypokalemia, hyponatremia, hypomagnesemia, hypercalcemia, hyperglycemia, hyperlipidemia, & hyperuricemia)

20) Banana peel:
Thiazide diuretics can cause a variety of adverse effects (e.g., **hypokalemia**, hyponatremia, hypomagnesemia, hypercalcemia, hyperglycemia, hyperlipidemia, and hyperuricemia)

21) Broken salt shaker:
Thiazide diuretics can cause a variety of adverse effects (e.g., hypokalemia, **hyponatremia**, hypomagnesemia, hypercalcemia, hyperglycemia, hyperlipidemia, & hyperuricemia)

22) Torn-up MAGazine:
Thiazide diuretics can cause a variety of adverse effects (e.g., hypokalemia, hyponatremia, **hypomagnesemia**, hypercalcemia, hyperglycemia, hyperlipidemia, & hyperuricemia)

23) Floating Calci-Yum ice cream:
Thiazide diuretics can cause a variety of adverse effects (e.g., hypokalemia, hyponatremia, hypomagnesemia, **hypercalcemia**, hyperglycemia, hyperlipidemia, & hyperuricemia)

24) Floating jar of candy:
Thiazide diuretics can cause a variety of adverse effects (e.g., hypokalemia, hyponatremia, hypomagnesemia, hypercalcemia, **hyperglycemia**, hyperlipidemia, & hyperuricemia)

25) Floating stick of butter:
Thiazide diuretics can cause a variety of adverse effects (e.g., hypokalemia, hyponatremia, hypomagnesemia, hypercalcemia, hyperglycemia, **hyperlipidemia**, & hyperuricemia)

26) Floating yellow yarn & needle:
Thiazide diuretics can cause a variety of adverse effects (e.g., hypokalemia, hyponatremia, hypomagnesemia, hypercalcemia, hyperglycemia, hyperlipidemia, & **hyperuricemia**)

27) Poker table with cracked Ace of spades:
ACE inhibitors (e.g., -pril) & **ARBs** (e.g., -sartan) are first-line treatment options for primary HTN

28(P) Coughing skeleton with braids:
ACE inhibitors can cause **cough due to increased bradykinin** → switch patient to ARB if they complain of ACE inhibitor-associated cough **(6)**

29) Banana daiquiri:
Side effects of ACE inhibitors & ARBs include **hyperkalemia** & angioedema

30) Swollen lip fish:
Side effects of ACE inhibitors & ARBs include hyperkalemia & **angioedema**

31) Calci-Yum soft serve machine:
Long-acting calcium channel blockers (e.g., amlodipine) are first-line treatment options for primary HTN

32) DIPPIN'S topping station:
Amlodipine is a long-acting dihydropyridine calcium channel blocker
33) Long lasting flavors!:
Amlodipine is a **long-acting** dihydropyridine calcium channel blocker

34) Dead sailor with swollen pant legs:
Lower extremity edema is the major side effect of amlodipine

35(P) Elevator Switch:
For patients in Stage 1 HTN whose BP is not controlled with your initial choice of pharmacotherapy, **switch to a different medication class (7)**

36(P) Gauge on deeper diver:
All patients with **Stage 2 hypertension** **(i.e., SBP ≥140 mmHg or DBP ≥90 mmHg)** require pharmacologic therapy **(8)**

37(P) Pill bottles surrounding deeper diver:
All patients with Stage 2 hypertension (i.e., SBP ≥140 mmHg or DBP ≥90 mmHg) **require pharmacologic therapy (8)**

38(P) Combination locker with 20/10 handles:
In patients with Stage 2 HTN, start **combination pharmacotherapy** (e.g., ACEi/ARB + CCB) **if SBP >20 above goal or DBP >10 above goal (9)**

39(P) 2 LARGE pill bottles:
If BP isn't controlled with initial combination therapy, **increase dosing** OR add a third agent **(10)**

40(P) 3 small pill bottles:
If BP isn't controlled with initial combination therapy, increase dosing OR **add a third agent (10)**

41(P) Black diver with pale thighs, reaching for Calci-Yum soft serve:
Thiazide diuretics OR calcium channel blockers are preferred for initial monotherapy in black patients with primary HTN **(11)**

42(P) ACE of spades poker table:
ACEis/ARBs are recommended in hypertensive patients with proteinuric CKD because they can slow the progression of kidney disease **(12)**

43(P) Rusty pulleys in murky protein water:
ACEis/ARBs are recommended in hypertensive patients **with proteinuric CKD** because they can slow the progression of kidney disease **(12)**

44(P) Diasweeties candy jar on poker table:
ACEis/ARBs are recommended in hypertensive patients **with diabetes & severe albuminuria (13)**
45(P) Floppy heart balloon:

ACEis/ARBs and Beta Blockers are recommended in hypertensive patients **with heart failure with reduced ejection fraction (aka HFrEF) (14)**

46(P) ACE of spades:
ACEis/ARBs and Beta Blockers are recommended in hypertensive patients with heart failure with reduced ejection fraction (aka HFrEF) **(14)**

47(P) Jack playing muted beta bugle:
ACEis/ARBs and **Beta Blockers** are recommended in hypertensive patients with heart failure with reduced ejection fraction (aka HFrEF) **(14)**

48) Heart-shaped harp with broken string:
In HFrEF patients with **history of ACS**, ACEis/ARBs & beta blockers have been shown to reduce mortality

49) Angel playing harp:
In HFrEF patients with history of ACS, ACEis/ARBs & beta blockers have been shown to **reduce mortality**

50) Fallen mineral crystal chandelier:
Aldosterone receptor antagonists (e.g., spironolactone, eplerenone) are used in certain HFrEF patients and also have some antihypertensive effects

51(P) Muted Beta bugle:
Beta blockers can help lower blood pressure in patients with conditions like AFib, angina, essential tremor, & migraine **(15)**

52(P) Irregularly irregular music:
Beta blockers can help lower blood pressure in patients with conditions like **AFib**, angina, essential tremor, & migraine **(15)**

53(P) ANVIL-shaped xylophone:
Beta blockers can help lower blood pressure in patients with conditions like AFib, **angina**, essential tremor, & migraine **(15)**

54(P) Shaking conductor's baton:
Beta blockers can help lower blood pressure in patients with conditions like AFib, angina, **essential tremor**, & migraine **(15)**

55(P) Head-shaped bell:
Beta blockers can help lower blood pressure in patients with conditions like AFib, CAD, essential tremor, & **migraine (15)**

56) Skeleton blowing long, loud party blower:
Adverse effects of beta blockers include **bronchoconstriction** & bradycardia

57) Broken heart-shaped pocket watch:
Side effects of beta-blockers include bronchoconstriction & **bradycardia**

58(P) Blown out alpha candle:
Alpha blockers may be helpful for hypertensive men with BPH **(16)**

59(P) Bloated diver with LOOPed tubing:
Loop diuretics (e.g., furosemide, bumetanide) are typically used to treat volume overload sx caused by HF or CKD, and have some antihypertensive effects **(17)**

60) Loud GONG:
Side effects of loop diuretics include **ototoxicity**, hypokalemia, & metabolic alkalosis

61) Falling banana peel:
Side effects of loop diuretics include ototoxicity, **hypokalemia**, & metabolic alkalosis

62) BASIC soap:
Side effects of loop diuretics include ototoxicity, hypokalemia, & **metabolic alkalosis**

63) Diver struggling with RESISTANT door:
Resistant hypertension = uncontrolled HTN despite a regimen of 3 or more drugs, including a diuretic

64) III bars on resistant door:
Resistant hypertension = uncontrolled HTN despite a regimen of **3 or more drugs, including a diuretic**

65(P) Spilled pills & bottles:
In a patient with resistant HTN, **first confirm that they are compliant with home meds** AND are prescribed an appropriate regimen **(18)**

66(P) Home med bag:
In a patient with resistant HTN, **first confirm that they** are compliant with home meds AND **are prescribed an appropriate regimen (18)**

67(P) Fallen crystal chandelier:
Aldosterone receptor antagonists (e.g., spironolactone, eplerenone)can be added for treatment of resistant HTN **(20)**

68) Banana wallpaper:
Side effects of aldosterone receptor antagonists include **hyperkalemia (both spironolactone & eplerenone)** and gynecomastia/ED (spironolactone only)

69) ==P== Passage to 2˚ deck:
Resistant HTN should raise suspicion for **SECONDARY** HTN (e.g., renal artery stenosis, renal parenchymal disease, primary aldosteronism, Cushing syndrome, pheochromocytoma, hyper/hypothyroidism, hyperparathyroidism, aortic coarctation) **(19)**

70) Round food dishes:
Side effects of aldosterone receptor antagonists include hyperkalemia (both spironolactone & eplerenone) and **gynecomastia/ED (spironolactone only)**

71(P) Muted beta bugle case:
Additional drugs used in the treatment of resistant HTN include **beta blockers (e.g., labetalol, carvedilol, bisoprolol)**, direct vasodilators (e.g., hydralazine, minoxidil), & clonidine **(21)**

72(P) Red shirt with dilated sleeves:
Additional drugs used in the treatment of resistant HTN include beta blockers (e.g., labetalol, carvedilol, bisoprolol), **direct vasodilators (e.g., hydralazine, minoxidil)**, & clonidine **(21)**

73(P) Bear rug with big CLAWs:
Additional drugs used in the treatment of resistant HTN include beta blockers (e.g., labetalol, carvedilol, bisoprolol), direct vasodilators (e.g., hydralazine, minoxidil), & **clonidine (21)**